

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE)

November 2013

CE402 Advanced Operating System

Date: 21.11.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following questions [07]

- (a) What is the main difference between a distributed operating system & a network operating system? Name two advantages of a microkernel over a monolithic kernel. [04]
- (b) What is the difference between a connection oriented and connectionless communication protocol? [03]

Q – 2 Answer the following questions [14]

- (a) Define: Remote Procedure Call (RPC). Explain basic operation of RPC. Explain Call Message & Reply Message format of RPC in brief. [05]
- (b) Explain future research trends in distributed file systems. [05]
- (c) Explain advantages and disadvantages of Distributed Systems over Independent PCs. [04]

OR

Q – 2 Answer the following questions [14]

- (a) List out the methods of dealing with the shared files in a distributed system. Explain any two of them in detail. [05]
- (b) Explain Remote Procedure Call (RPC) mechanism in detail. Which transport layer protocol is used to implement RPC? Justify it. [05]
- (c) A file is replicated on 10 servers. List all the combinations of read quorum and write quorum that are permitted by the voting algorithm. [04]

Q – 3 Answer the following questions [14]

- (a) What do you mean by Cache Consistency? Imagine that a distributed file uses client caching with a delayed write back policy. One machine opens, modifies, and closes a file. About half a minute later, another machine reads the file from the server. Which version does it get? [04]
- (b) Explain distributed file system implementation in detail. [05]

OR

- (b) Explain stateless Server in detail. Why stateless servers are more useful in Distributed System? Why do stateless servers have to include a file offset in each request? Is this also needed for stateful servers? [05]
- (c) Explain distributed file system principles in detail. [05]

SECTION – II

Q – 4 Answer the following questions [07]

- (a) List basic goals of Distributed system and explain any two of them in detail. [04]
- (b) What are the advantages of allocating objects such as proc structures and descriptor table blocks dynamically? What are the drawbacks? [03]

Q – 5 Answer the following questions [14]

- (a) Briefly explain the differences between user mode and kernel mode operation. Why two modes of operations are required in UNIX? [04]
- (b) What resources are used when a thread created? How do they differ from those when a process is created? [05]
- (c) Which are the properties of NUMA multiprocessors? Explain the observation of NUMA algorithms in brief. [05]

OR

Q – 5 Answer the following questions [14]

- (a) What is shared memory? Compare bus-based, ring-based and switched multiprocessor. [04]
- (b) Explain Berkeley algorithm of clock synchronization in detail. [05]
- (c) Explain in detail about PCB and how process represented in system memory. [05]

Q – 6 Answer the following questions [14]

- (a) Which are the two basic approaches for distributed mutual exclusion? Explain any one approach in detail. [07]
- (b) Explain different type component faults. And also explain various approaches to make a distributed system fault tolerant. [07]

OR

Q – 6 Answer the following questions [14]

- (a) Explain various consistency models of distributed shared memory in brief. [07]
- (b) Explain how scheduling is done in distributed systems with an example. [07]